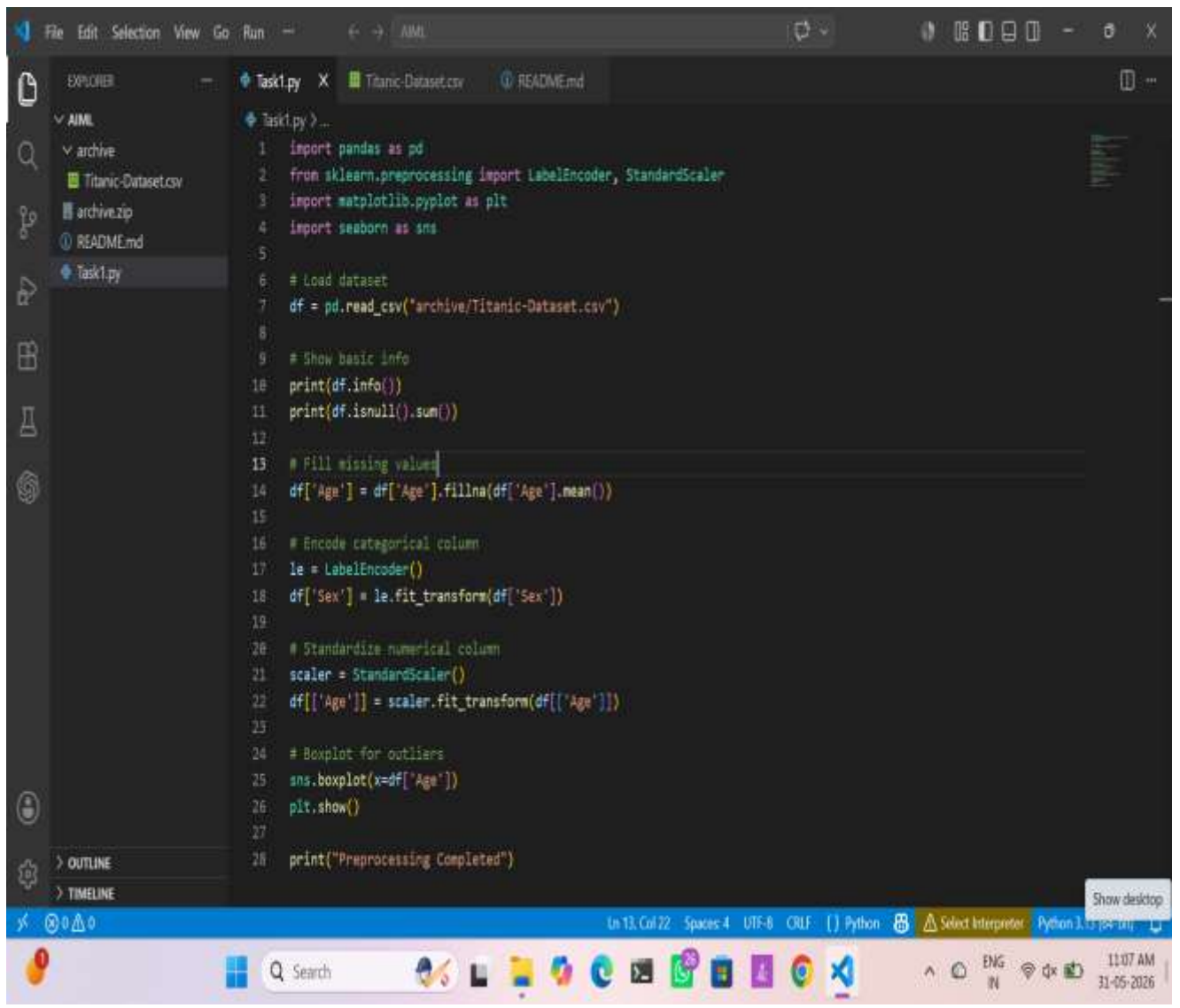


Task 1



The screenshot displays a Jupyter Notebook environment with a dark theme. The left sidebar shows the file explorer with a project named 'A1ML' containing an 'archive' folder with 'Titanic-Dataset.csv', 'archive.zip', and 'README.md', and a 'Task1.py' file. The main editor area shows the 'Task1.py' file with the following Python code:

```
1 import pandas as pd
2 from sklearn.preprocessing import LabelEncoder, StandardScaler
3 import matplotlib.pyplot as plt
4 import seaborn as sns
5
6 # Load dataset
7 df = pd.read_csv("archive/Titanic-Dataset.csv")
8
9 # Show basic info
10 print(df.info())
11 print(df.isnull().sum())
12
13 # Fill missing values
14 df['Age'] = df['Age'].fillna(df['Age'].mean())
15
16 # Encode categorical column
17 le = LabelEncoder()
18 df['Sex'] = le.fit_transform(df['Sex'])
19
20 # Standardize numerical column
21 scaler = StandardScaler()
22 df[['Age']] = scaler.fit_transform(df[['Age']])
23
24 # Boxplot for outliers
25 sns.boxplot(x=df["Age"])
26 plt.show()
27
28 print("Preprocessing Completed")
```

The bottom status bar indicates the current position is 'Ln 13, Col 22' with 'Spaces 4' and 'UTF-8' encoding. It also shows the Python interpreter is set to 'Python 3.11 [base]'. The Windows taskbar at the very bottom shows the search bar, task view button, and several application icons, with the system clock displaying '11:07 AM 31-05-2026'.

Output :-

The screenshot shows a Jupyter Notebook environment. The left sidebar contains a file explorer with folders like 'archive' and files like 'Task1.py'. The main area displays the output of a Python script that loads a dataset and prints its summary statistics.

```
PS C:\Users\ONGAI\OneDrive\Desktop\ADML> python -u "c:\users\ONGAI\OneDrive\Desktop\ADML\Task1.py"
```

```
<class 'pandas.DataFrame'>
```

```
RangeIndex: 891 entries, 0 to 890
```

```
Data columns (total 12 columns):
```

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	str
4	Sex	891 non-null	str
5	Age	714 non-null	float64
6	SibSp	891 non-null	int64
7	Parch	891 non-null	int64
8	Ticket	891 non-null	str
9	Fare	891 non-null	float64
10	Cabin	204 non-null	str
11	Embarked	889 non-null	str

```
dtypes: float64(2), int64(5), str(5)
```

```
memory usage: 83.7 KB
```

```
None
```

```
PassengerId    0
```

```
Survived        0
```

```
Pclass          0
```

```
Name           0
```

```
Sex             0
```

```
Age            177
```

```
SibSp           0
```

```
Parch          0
```

```
Ticket         0
```

```
Fare           0
```

```
Cabin         687
```

```
Embarked        2
```

The bottom status bar indicates the system is running Linux GUI, has 4 spaces, uses UTF-8 encoding, and is running Python 3.11.04 bit.

```

File Edit Selection View Go Run - - - - -
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python Debug Console + - - - - -

Entered 2
dtype: int64
Preprocessing Completed
PS C:\Users\QMSAI\OneDrive\Desktop\ADML> ^C
PS C:\Users\QMSAI\OneDrive\Desktop\ADML>
PS C:\Users\QMSAI\OneDrive\Desktop\ADML> c++ cd "x:\Users\QMSAI\OneDrive\Desktop\ADML" & "x:\Users\QMSAI\AppData\Local\Programs\Python\Python311\python.exe"
" % (Users\QMSAI\vscode\extensions\ms-python.debugpy-2023.8.6-win32-x64\bundle\libs\debugpy\launcher" "55311" "--" "x:\Users\QMSAI\OneDrive\Desktop\ADML\T
ask1.py"
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column      Non-Null Count  Dtype  
---  --
 0   PassengerId  891 non-null    int64  
 1   Survived    891 non-null    int64  
 2   Pclass      891 non-null    int64  
 3   Name        891 non-null    str     
 4   Sex         891 non-null    str     
 5   Age         714 non-null    float64 
 6   SibSp       891 non-null    int64  
 7   Parch       891 non-null    int64  
 8   Ticket      891 non-null    str     
 9   Fare        891 non-null    float64 
10   Cabin       294 non-null    str     
11   Embarked    889 non-null    str     

dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
None
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
locking completed.
In 13. Col 32 - Spaces 4 UTF-8 CRLF Python 3.11.2 Python 3.11.64 M

```

Graph :-

